

FINAL EXPLANATION: MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: MATHEMATICS GRADE 10: AREA OF PART OF A CIRCLE (SUB-STRAND 2.3)

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: How can we use the radius and the central angle of a circular pizza slice to calculate its exact area, and apply the same method to solve real problems in farming, design, and everyday life in Kenya? Answer every section in the space under each prompt. Write full sentences for explanations. For calculations, show every step: write the formula, substitute the values, then calculate. Use the π button on your calculator and do not round π early. Give final answers to 2 decimal places with correct units. Remember the pizza (or chapati) cut into slices of about 30° , 45° , 60° , 90° and 135° from Lesson 1, and the circular shamba in Nakuru from Lesson 10.

Section 1: Why do wider slices have larger areas?

Think about the pizza (or chapati) that was cut into unequal slices, all with the same radius. Look at the 45° slice and the 90° slice. (a) What fraction of the whole circle is each slice? (b) Without calculating any area, explain how the area of the 90° slice compares with the area of the 45° slice, and why. (c) Name the relationship between central angle and sector area when the radius stays fixed.

(a) A full circle has a central angle of 360° . The 45° slice is $45/360 = 1/8$ of the circle. The 90° slice is $90/360 = 1/4$ of the circle. (b) The area of a sector is the fraction $\theta/360^\circ$ multiplied by the area of the whole circle, πr^2 . Both slices come from circles of the same radius, so πr^2 is the same for both. The 90° slice has twice the fraction ($1/4$ is twice $1/8$), so its area is exactly twice the area of the 45° slice. (c) The relationship is direct proportion. When the radius is fixed, doubling the central angle exactly doubles the sector area. This is why the data points for angle against sector area formed a straight line through the origin in our sector-card activity.

Section 2: Area and arc length of a pizza slice

A circular pizza has a radius of 28 cm. A slice is cut with a central angle of 135° . (a) Write the formula for the area of a sector. (b) Calculate the area of the slice. (c) Calculate the length of the curved crust edge (the arc) of this slice. (d)

(a) Area of a sector $A = (\theta/360^\circ) \times \pi r^2$, where θ is the central angle in degrees and r is the radius. (b) $A = (135/360) \times \pi \times 28^2 = 0.375 \times \pi \times 784 = 0.375 \times 2463.01 = 923.63 \text{ cm}^2$. (c) Arc length $L = (\theta/360^\circ) \times 2\pi r = (135/360) \times 2 \times \pi \times 28 = 0.375 \times 175.93 = 65.97 \text{ cm}$. (d) The slice takes up $135/360$ of the whole circle, so it has that same fraction of the circle's area and of the circle's circumference.

Explain in one sentence why the fraction you used is the same in (b) and (c).	
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Section 3: The circular shamba in Nakuru

A circular shamba in Nakuru has a radius of 21 m. An 80° sector of it is fenced. The rest of the shamba, which has a central angle of 280°, is not fenced. (a) Calculate the length of the arc along the outer edge of the fenced sector. (b) Calculate the area of the fenced (minor) sector. (c) Calculate the area of the unfenced (major) sector in two different ways, and show that both methods agree.	(a) $L = (80/360) \times 2 \times \pi \times 21 = (2/9) \times 131.95 = 29.32$ m. (b) $A = (80/360) \times \pi \times 21^2 = (2/9) \times 1385.44 = 307.88$ m². (c) Method 1: Use the formula with $\theta = 280^\circ$. $A = (280/360) \times \pi \times 21^2 = (7/9) \times 1385.44 = 1077.57$ m² (to 2 d.p.). Method 2: Subtract the minor sector from the whole circle, keeping the unrounded values: $\pi \times 21^2 = 1385.4424$ m² and the minor sector = $(2/9) \times 1385.4424 = 307.8761$ m², so major sector = $1385.4424 - 307.8761 = 1077.5663 = 1077.57$ m² (to 2 d.p.). Both methods give 1077.57 m². The formula works for any angle, minor or major, because $\theta/360^\circ$ is always the fraction of the whole circle.
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Section 4: Segments of the shamba

The 80° sector in the Nakuru shamba (radius 21 m) has its two radii joined by a straight chord. This chord cuts the circle into a minor segment and a major segment. (a) Calculate the area of the triangle formed by the two radii and the chord. (b) Calculate the area of the minor segment. (c) Calculate the area of the major segment. (d) Explain why the minor segment is smaller than the 80° sector.	(a) Triangle area = $\frac{1}{2}r^2 \sin \theta = \frac{1}{2} \times 21^2 \times \sin 80^\circ = 220.5 \times 0.9848 = 217.15$ m². (b) Minor segment = sector area – triangle area = $307.88 - 217.15 = 90.73$ m². (c) Major segment = area of whole circle – minor segment = $1385.44 - 90.73 = 1294.71$ m². (d) The sector includes the triangle as well as the curved region between the chord and the arc. The segment is only the curved region, so the triangle area, which is positive, is subtracted. This makes the segment smaller than the sector.
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Section 5: Spot the error and answer the driving question

A learner works on the 80° sector of the shamba with radius 21 m. They write: 'Area	(a) The first learner calculated the sector area, not the segment area. They forgot to subtract the triangle area $\frac{1}{2}r^2 \sin \theta$. The correct method is segment = sector – triangle = $307.88 - 217.15 = 90.73$ m². (b) The second learner used 42 m, the diameter, instead of the radius. Since r is squared, the answer becomes 4 times too large. The correct value of r is 21 m, half of 42 m. (c)
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of the minor segment = $(80/360) \times \pi \times 21^2 = 307.88 \text{ m}^2$. Another learner works on the same sector and writes: 'Area = $(80/360) \times \pi \times 42^2$.' (a) What mistake did the first learner make, and what is the correct method? (b) What mistake did the second learner make, and how does it change the answer? (c) Using what you learned from the pizza, the shamba, and the segments, write a clear answer to the driving question in 3 to 4 sentences. Include one real Kenyan situation where this method is useful.	Any slice of a circle is a fraction $\theta/360^\circ$ of the whole circle, so if we know only the radius and the central angle we can find its exact area using $A = (\theta/360^\circ) \times \pi r^2$. Doubling the angle doubles the area when the radius is fixed. Arc length uses the same fraction of $2\pi r$. For a segment, we subtract the triangle area $\frac{1}{2}r^2 \sin \theta$ from the sector area. For example, a farmer in Nakuru can use this to work out the exact area of a fenced sector of a circular shamba, or a seller can price watermelon slices fairly.
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining proportional reasoning	Clearly links central angle to the fraction $\theta/360^\circ$ and explains why doubling the angle exactly doubles the area when the radius is fixed, referring to the pizza slices.	States that a larger angle gives a larger area and mentions direct proportion, but the link to the fraction of 360° is incomplete.	Says only that bigger slices are bigger, with no fraction or proportion reasoning.
Using formulas for arc length and sector area (minor and major)	Chooses the correct formulas, substitutes correctly, uses the radius, and gives accurate answers to 2 decimal places with units.	Uses the correct formulas with minor slips in substitution, rounding or units.	Uses the wrong formula or the diameter for the radius, or gives answers with no working.
Calculating segment areas	Correctly finds the triangle area, subtracts it from the sector to get the minor segment, and subtracts the minor segment from the whole circle to get the major segment.	Completes most steps correctly but makes a calculation slip or gives a weak explanation of why the triangle is subtracted.	Does not subtract the triangle, or confuses the segment with the sector.
Diagnosing errors and communicating the answer to the driving question	Names each error precisely, corrects it, and gives a clear, complete answer to the driving question with a relevant Kenyan example.	Identifies the errors and gives a reasonable answer to the driving question, but with limited detail or no clear example.	Does not identify the errors, or gives an answer unrelated to the formulas learned.

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